Predictive Modeling of Alzheimer's Disease

Comparison of MLP and Random Forest Classifiers

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Exploratory Data Analysis

1. Dataset

The dataset contains **2,149 observations** and **14 variables** describing patients, including age, BMI, physical activity, and the occurrence of cognitive problems. The data has no missing values and no duplicates. The target variable is **Diagnosis**, where 0 indicates the absence of Alzheimer's disease and 1 indicates its presence.

Histograms revealed that most patients are between 70-85 years old, and binary variables (e.g., MemoryComplaints, BehavioralProblems) are imbalanced. Correlation analysis showed that the strongest linear associations with Diagnosis come from ADL, MemoryComplaints, and BehavioralProblems.

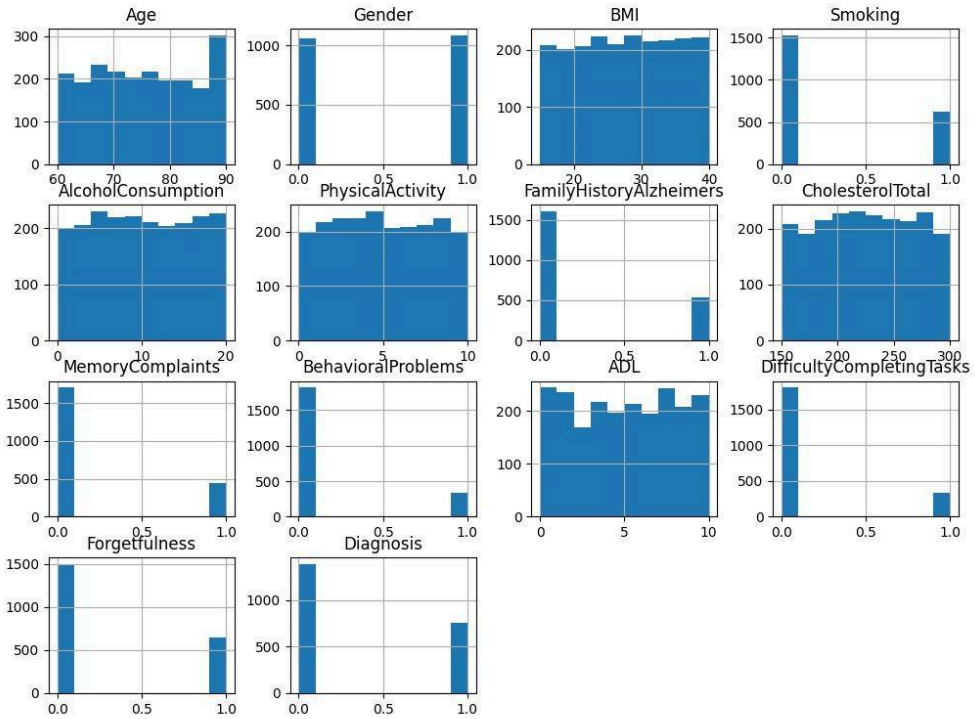

Figure 1. Histograms of all features in the dataset.

2. Feature Selection

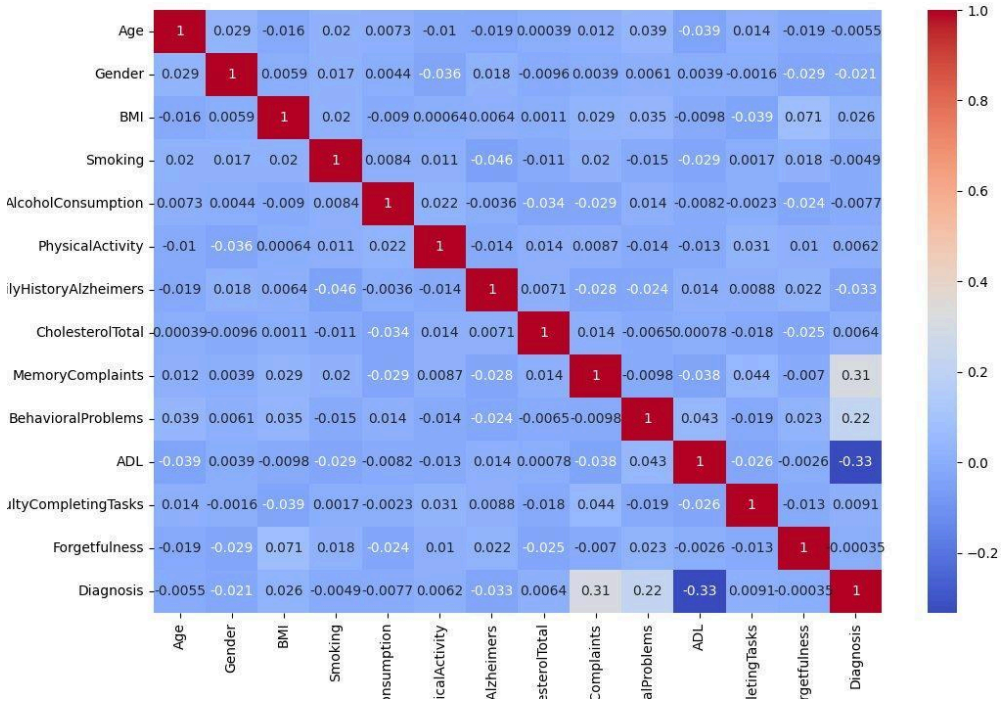

Figure 2. Correlation heatmap. The bottom row shows correlation with the target (Diagnosis).

Selected predictors

Feature	Justification
Age	Strong risk factor for Alzheimer's; notable correlation with the target.
MemoryComplaints	Reported memory issues are one of the earliest disease symptoms (corr. 0.31).
BehavioralProblems	Behavioral changes often accompany neurodegenerative diseases (corr. 0.22).
ADL	Lower scores indicate difficulty with daily activities (corr. -0.33).
FamilyHistoryAlzheimers	Genetic factor increasing disease risk. Low statistical correlation, but strong medical justification.
DifficultyCompletingTasks	Cognitive impairment indicator. Low statistical correlation, but medically relevant.

Excluded features

Gender, BMI, Smoking, AlcoholConsumption, PhysicalActivity, CholesterolTotal, and Forgetfulness were excluded from the model due to negligible correlation with the target variable. Their inclusion was not expected to improve model performance.

3. Models

MLP (Multi-Layer Perceptron)

Parameter	Value
Architecture	Two hidden layers (20, 10 neurons)
Activation function	ReLU
Maximum iterations	500
Random state	seed (derived from dataset indices)
Preprocessing	StandardScaler (fit on train, transform on test)

Random Forest

Parameter	Value
Number of trees	300
max_depth	None (unrestricted)
Random state	seed
n_jobs	-1 (all CPU cores)

The dataset was split 70/30 into training and test sets, with the random seed derived from the integer mean of the dataset indices for reproducibility.

4. Results

Test set metrics

Metric	MLP	Random Forest
Accuracy	0.72	0.68
Specificity	0.86	0.77
Sensitivity (Recall)	0.48	0.53
F1-score	0.56	0.55
AUC	**0.776**	**0.748**

ROC curves

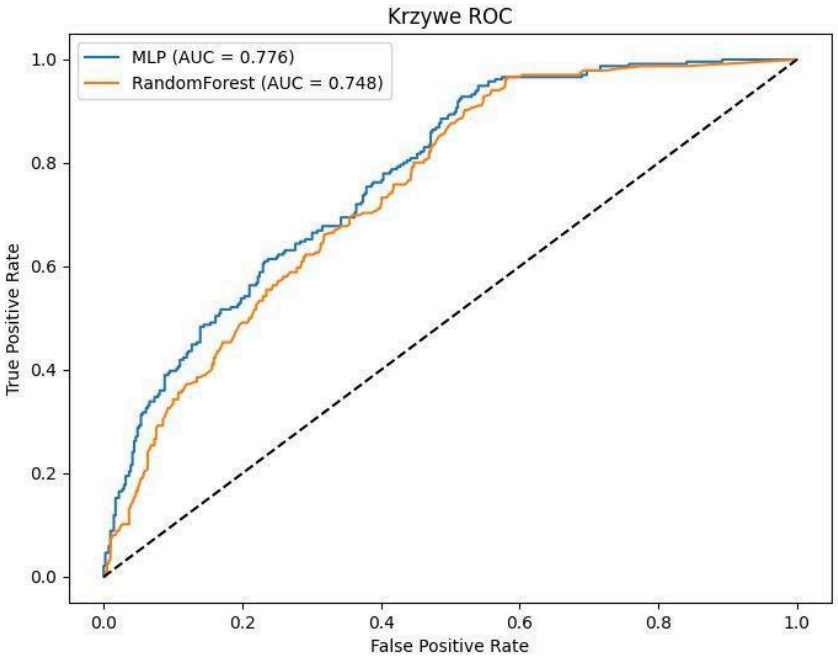

Figure 3. ROC curves for both models on the test set.

5. Model Comparison and Selection

Both MLP and Random Forest models successfully learn dependencies in the data and can predict Alzheimer's diagnosis. However, they differ significantly in how they generalize to unseen data.

Random Forest

Random Forest achieved nearly perfect results on the training set (accuracy ~ 0.998), meaning it essentially memorized the training data. On the test set, however, its accuracy dropped to ~0.68. This large gap clearly indicates **overfitting** - the model learned specific details of the training set rather than general patterns.

MLP

MLP achieved lower training accuracy, but its test set results are more stable (accuracy ~ 0.72). This means the neural network **generalizes better** and handles previously unseen data more reliably.

Additionally, the AUC for MLP was **0.776**, compared to **0.748** for Random Forest. MLP therefore better distinguishes between sick and healthy individuals across the entire range of decision thresholds, which is also visible on the ROC curve.

6. Conclusion

The MLP model is the better choice for the following reasons:

- Higher accuracy on test data, which truly reflects model quality.
- Better generalization with no strong overfitting.
- Higher AUC, indicating better class separation.

Random Forest, despite excellent results on the training set, is too closely fitted to the training data, which limits its practical usefulness.

Limitations and future work

- **Low sensitivity (0.48-0.53)** in both models - too many false negatives. In a medical context, this means missed diagnoses. Possible remedies: decision threshold tuning, class balancing (SMOTE), or cost-sensitive learning.
- **Hyperparameter tuning** using GridSearchCV could improve both models.
- **Random Forest overfitting** could be reduced by limiting max_depth, increasing min_samples_leaf, or applying feature subsampling.